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VMD–IMF Enhanced Hyper Graph Attention Module Based Reinforcement Learning for Portfolio Optimization

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Abstract

Stock portfolio optimization is an important area in financial decision-making. It focuses on optimizing the balance between risk and return. Our approach is centered exclusively on stock data, based on the assumption that market dynamics and price movements capture all relevant available information.

We take advantage of the non-stationary and heterogeneous nature of financial time series by decomposing features into multiple intrinsic modes, which provide a richer representation of the temporal information and integrate with the hyper-graphs and attention mechanisms which represent complex relationships within the stocks in the portfolio.

This paper presents a novel hybrid model that integrates Intrinsic Mode Functions (IMFs)

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Notes

1. <https://pypi.org/project/vmdpy/>.

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